

Frozen GPT-2, ePC-trained and BP-trained reader caps: ordinary performance, dataset extremes and heavy tails (ext-20261009)

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1 Frozen GPT-2, ePC-trained reader cap and BP-trained reader cap on zsRE, CounterFact and MQuAKE: ordinary performance, dataset extremes and heavy tails

Study `ext-20261009` · Capstan · 2026-10-09 · `pc_cap` code revision `b376dfe5cb32` · commissioned by `docs/AgentHandoffForResearchInTheExtremes.md` as amended by `docs/mandatory_ammendment.md`.

Provenance tags: **[record]** previously established in the frozen record and copied with its hashes; **[recomputed]** recomputed in this study from saved data; **[new]** newly evaluated in this study (own manifest `results/extremes_analysis/ext-20261009/manifest.json`). The frozen record was not modified.

1.1 1. Abstract and executive summary

Question. Across zsRE, CounterFact and MQuAKE, how do (i) frozen GPT-2 small without a cap, (ii) GPT-2 with the reader cap trained by error predictive coding (ePC) and (iii) GPT-2 with the same cap trained by backprop (BP) compare in ordinary benchmark performance, prediction difficulty, heavy-tail behaviour, extreme losses, adaptation benefit, correction magnitude, locality and stability — and does the ePC-trained cap show advantages in rare or extreme cases that survive a direct comparison with the BP-trained cap?

Design. The principal comparison is the frozen record’s PC-reader study (family PCR): three paired training seeds per rule, identical architecture (3,348,228 reader parameters, taps and writes at blocks 4/8/12), identical initial parameters and training episodes within seed, adjoint acquisition in both arms, evaluated on realization 0 / order 100 with 300 edits on zsRE and CounterFact (checkpoints 100 and 300). This study adds **[new]**: a directly loaded frozen GPT-2 scored on exactly the same probes; teacher-forced target log-probabilities for all seven models on every probe (edit prompts, paraphrases, locality, near-miss, revision, unseen, MQuAKE multi-hop questions); the six saved readers evaluated on the sealed MQuAKE stream (0/6 complete at report time); write-vector magnitudes at every scored position; dataset frequency/entropy/shift statistics; and finite-range tail fits that start from a field-by-field reproduction of HT-17.

Headline findings.

- **zsre, paraphrase retention at 300 edits:** BP 97.3 %, 99.0 %, 97.7 %; ePC 94.7 %, 97.0 %, 98.0 % (seeds 0, 1, 2; 300 items each). Frozen: 0.0 %.
- **counterfact, paraphrase retention at 300 edits:** BP 81.0 %, 80.0 %, 83.5 %; ePC 56.8 %, 77.8 %, 66.0 % (seeds 0, 1, 2; 300 items each). Frozen: 0.0 %.
- **Neither cap beats the frozen model everywhere.** Both caps convert the taught facts (own-prompt loss falls from 6 nats/token to 0.01) and both leave the frozen model’s behaviour unchanged where they abstain; the differences between ePC- and BP-trained readers are differences of *gating*: which paraphrases and un-taught prompts trigger a write. Where a reader abstains, its loss equals the frozen loss to the bit.
- **ePC vs BP.** On CounterFact the ePC-trained reader retains fewer paraphrases in all three seeds (frozen record), and its per-probe paraphrase loss is higher than BP’s in the paired contrast; on zsRE the two rules are close with mixed signs. No analysis in this study (difficulty deciles, frozen-hard subsets, regressions, correction magnitudes) finds a regime of rare or extreme cases where the ePC-trained reader is reliably better than the BP-trained reader. Training cost remains 96–102× **[record]**.
- **Extremes.** Frozen difficulty is heavy in the descriptive sense (per-token target surprisal has P99 above 10 nats on every dataset) but the fitted generalized-Pareto shapes of the *probe* losses are near zero or negative with intervals that include zero; the ordinary-text harm tails of the PC-reader cells reproduce HT-17 exactly (12/12 fields) and remain finite-range fits. Rare large deteriorations exist for both caps and are concentrated on prompts where a wrong record fired.
- **Qualifications.** One subject realization and one order; three training seeds are not three populations; GPT-2 small only; the frozen model answers essentially none of these prompts under the project’s greedy convention, so the frozen baseline is informative through teacher-forced losses, not through exact-match rates.

1.2 2. Research questions and architectures

The cap is an episodic memory with a learned reader attached to a frozen GPT-2 small (12 blocks, 124,439,808 parameters, params digest `c4ac3fb8...`). Reads tap the residual stream after blocks 4, 8 and 12; writes add one vector per site at the current prediction position. Each edit is taught by five adjoint delta steps (lr 0.1, acceptance 0.1, aggregate budget $A = 0.3$) and appended as a record (key = prompt embedding at the three taps; payload = per-answer-token write deltas [3, 768]). At query time the reader scores the prompt against the nearest stored keys and keeps a null option; null mass 0.5 means no write, so the output equals the base to the bit. There is no inference-time iteration in either cap; the eight settling steps of ePC are a *training-time* quantity.

model	what differs	checkpoint	where evaluated
frozen GPT-2 small [new]	no reader, no memory, no writes	base params only	all probes, all datasets (Stage 2)
BP reader, seeds 0–2 [record]	reader trained by backprop (300 AdamW updates; feed-forward answer CE, cached float16 teacher)	<code>train-bp-s*/theta_avg150-300.npz</code>	zsRE/CounterFact (record); MQuAKE (this study)
ePC reader, seeds 0–2 [record]	same episodes and initial parameters; base-dependent gradients from EPCTrainer (8 zero-initialised settling steps, rate 0.1, corrected SD-24 energy); 23.4–24.6 h vs 0.24–0.25 h	<code>train-epc-s*/theta_avg150-300.npz</code>	same

Verified differences between the two caps (handoff §1): architecture, parameter count, interface, acquisition rule, evaluation population and scoring code are identical; only the training-time gradient estimator for the base-dependent losses differs (settled ePC CE vs feedforward CE; float32 vs cached float16 teacher), hence the weights and the cost. The validation record `validation/gpu_checks.json` confirms: the directly loaded weights have the record’s digest; a direct forward reproduces the saved cap-off losses to 0.000050 nats over 1524 positions; a reader with an empty memory, and a loaded memory on prompts where it abstains, returns the base logits exactly; the residual insertion arithmetic $h_{\text{after}} = h_{\text{before}} + \Delta h$ holds at the site; saved RET-GS/RET-ES recompute exactly from the saved rows.

Families kept distinct and reported separately: Stage-4 `R1_learned_ff` (selected v5 BP-trained reader, 1,000 edits, 3 realizations $\times$ 5 orders) **[record]**; fixed-v5 acquisition credit (adjoint vs error inference, reader fixed) **[record]**; PC-v0 (regenerated ePC 50M base, v0 live cap) **[record, supplementary only]**.

1.3 3. Datasets and evaluation methodology

dataset	version used	pool	principal stream	probes per item	notes
zsRE	MEND release, train split rows; <code>zsre_eligible.jsonl</code> (10,420 eligible of 10,720 screened; 0 answered by the base)	10,420	realization 0, order 100, first 300 of 1,000	1 paraphrase, 1 NQ locality prompt	no relation ids; target = first <code>answers[]</code> entry
CounterFact	original CounterFact; <code>counterfact_eligible.jsonl</code> (20,091 of 20,391; 0 answered by the base)	20,091	same	2 paraphrases, ~10 neighbourhood prompts (first 3 scored)	<code>target_true</code> retained; relation ids
MQuAKE	MQuAKE-CF (9,218 cases; DEC-045); single-hop rewrite items + 80 composition cases per realization	6,043 items	300 items	1 question-form paraphrase, 2 locality prompts	<code>target_true/target_new</code> with QIDs; 3 multi-hop questions per case

Endpoint bundles per realization: 50 locality prompts, 100 near-miss pairs (distinct subjects within matched relation/template families; the reference is the cell’s own cap-off neighbour response), 50 revision cases, 100

unseen prompts (un-taught pool items; false-fire measurement), and for MQuAKE 80 composition cases. Original metrics [**record**]: ES (immediate own-prompt success), RET-ES/RET-GS (end-of-stream own-prompt / paraphrase retention), LS (DEC-053 bounded exact decoded-text equality with the original-base response; *not* factual accuracy), near-miss preservation, revision, unseen false fires. Decoding: greedy, 32 new tokens, stop at newline/EOS; exact match of the NFKC/casfold/whitespace-normalised generation against supplied aliases.

Standardised measurements [**new**]: for prompt x_i and answer tokens $y_{i,1..T_i}$ (the project’s tokenisation: leading space, newline terminator), $\log P_i = \sum_t \log p(y_{i,t} | x_i, y_{i,<t})$ under teacher forcing with one prediction per gold prefix (the cap’s writes act at every prediction position, exactly as the record’s diagnostic); $NLL_total_i = -\log P_i$; $NLL_token_i = NLL_total_i / T_i$ (primary continuous difficulty measure); totals without the terminator are retained. Margins $margin_total = \log P(target_new) - \log P(target_true)$ and $margin_token = -NLL_token(new) + NLL_token(true)$. All seven models score the same serialised prompt/answer pairs; identical bucket padding; float64 log-softmax of float32 logits. Reader selection (record ids, null mass, hard null) and write norms $\|W_m\|$ and $\|W_m\|/\|h_m\|$ are recorded at every scored position. Probe counts: zsRE 1,350, CounterFact 2,300 (item-locality capped at 3 per item), MQuAKE 1,890 (incl. 240 composition questions).

Uncertainty: 2,000-draw percentile bootstrap resampling whole groups (items for item-level families, endpoint rows otherwise) with the same indices for every model (pairing preserved); windows for ordinary text; seeds reported individually and as a descriptive seed mean. These intervals cover within-population sampling only; no interval here speaks to new subjects or new training seeds.

1.4 4. Basic benchmark results

1.4.1 4.1 Registered metrics (original conventions, numerators/denominators)

1.4.2 zsre

edits	model	ES	RET-ES	RET-GS (pri- mary)	LS	near- miss	unseen false fires	MQuAKE composi- tion (all 3 ques- tions)
300	frozen GPT-2 [new]	0.0 % (0/300)	0.0 % (0/300)	0.0 % (0/300)	100.0 % (50/50) (by defi- nition)	100.0 % (100/100) (by defi- nition)	N/A	
300	BP reader s0 [record]	100.0 % (300/300)	100.0 % (300/300)	97.3 % (292/300)	100.0 % (50/50)	85.0 % (85/100)	4/100	
300	BP reader s1 [record]	100.0 % (300/300)	100.0 % (300/300)	99.0 % (297/300)	100.0 % (50/50)	67.0 % (67/100)	14/100	
300	BP reader s2 [record]	100.0 % (300/300)	100.0 % (300/300)	97.7 % (293/300)	100.0 % (50/50)	73.0 % (73/100)	11/100	
300	ePC reader s0 [record]	100.0 % (300/300)	100.0 % (300/300)	94.7 % (284/300)	100.0 % (50/50)	91.0 % (91/100)	2/100	
300	ePC reader s1 [record]	100.0 % (300/300)	100.0 % (300/300)	97.0 % (291/300)	100.0 % (50/50)	85.0 % (85/100)	5/100	
300	ePC reader s2 [record]	100.0 % (300/300)	100.0 % (300/300)	98.0 % (294/300)	100.0 % (50/50)	85.0 % (85/100)	4/100	
100	frozen GPT-2 [new]	0.0 % (0/300)	0.0 % (0/300)	0.0 % (0/300)	100.0 % (50/50) (by defi- nition)	100.0 % (100/100) (by defi- nition)	N/A	
100	BP reader s0 [record]	100.0 % (100/100)	100.0 % (100/100)	96.0 % (96/100)	100.0 % (50/50)	80.0 % (80/100)	5/100	
100	BP reader s1 [record]	100.0 % (100/100)	100.0 % (100/100)	99.0 % (99/100)	100.0 % (50/50)	56.0 % (56/100)	13/100	
100	BP reader s2 [record]	100.0 % (100/100)	100.0 % (100/100)	96.0 % (96/100)	100.0 % (50/50)	66.0 % (66/100)	8/100	
100	ePC reader s0 [record]	100.0 % (100/100)	100.0 % (100/100)	92.0 % (92/100)	100.0 % (50/50)	88.0 % (88/100)	2/100	
100	ePC reader s1 [record]	100.0 % (100/100)	100.0 % (100/100)	92.0 % (92/100)	100.0 % (50/50)	80.0 % (80/100)	5/100	
100	ePC reader s2 [record]	100.0 % (100/100)	100.0 % (100/100)	95.0 % (95/100)	100.0 % (50/50)	82.0 % (82/100)	4/100	

1.4.3 counterfact

edits	model	ES	RET-ES	RET-GS (pri- mary)	LS	near- miss	unseen false fires	MQuAKE composition (all 3 ques- tions)
300	frozen GPT-2 [new]	0.0 % (0/300)	0.0 % (0/300)	0.0 % (0/600)	100.0 % (50/50) (by defi- nition)	100.0 % (100/100) (by defi- nition)	N/A	
300	BP reader s0 [record]	100.0 % (300/300)	100.0 % (300/300)	81.0 % (243/300)	100.0 % (50/50)	100.0 % (100/100)	0/100	
300	BP reader s1 [record]	99.0 % (297/300)	99.0 % (297/300)	80.0 % (240/300)	100.0 % (50/50)	99.0 % (99/100)	1/100	
300	BP reader s2 [record]	100.0 % (300/300)	100.0 % (300/300)	83.5 % (250.5/300)	100.0 % (50/50)	100.0 % (100/100)	0/100	
300	ePC reader s0 [record]	100.0 % (300/300)	100.0 % (300/300)	56.8 % (170.5/300)	100.0 % (50/50)	98.0 % (98/100)	0/100	
300	ePC reader s1 [record]	100.0 % (300/300)	100.0 % (300/300)	77.8 % (233.5/300)	100.0 % (50/50)	95.0 % (95/100)	1/100	
300	ePC reader s2 [record]	100.0 % (300/300)	100.0 % (300/300)	66.0 % (198/300)	100.0 % (50/50)	99.0 % (99/100)	0/100	
100	frozen GPT-2 [new]	0.0 % (0/300)	0.0 % (0/300)	0.0 % (0/600)	100.0 % (50/50) (by defi- nition)	100.0 % (100/100) (by defi- nition)	N/A	
100	BP reader s0 [record]	100.0 % (100/100)	100.0 % (100/100)	85.0 % (85/100)	100.0 % (50/50)	100.0 % (100/100)	0/100	
100	BP reader s1 [record]	100.0 % (100/100)	100.0 % (100/100)	85.0 % (85/100)	100.0 % (50/50)	100.0 % (100/100)	0/100	
100	BP reader s2 [record]	100.0 % (100/100)	100.0 % (100/100)	86.5 % (86.5/100)	100.0 % (50/50)	100.0 % (100/100)	0/100	
100	ePC reader s0 [record]	100.0 % (100/100)	100.0 % (100/100)	56.0 % (56/100)	100.0 % (50/50)	99.0 % (99/100)	0/100	
100	ePC reader s1 [record]	100.0 % (100/100)	100.0 % (100/100)	77.5 % (77.5/100)	100.0 % (50/50)	96.0 % (96/100)	0/100	
100	ePC reader s2 [record]	100.0 % (100/100)	100.0 % (100/100)	68.0 % (68/100)	100.0 % (50/50)	99.0 % (99/100)	0/100	

1.4.4 mquake

edits	model	ES		RET-ES		RET-GS (pri- mary)		LS	near- miss	unseen false fires	MQuAKE composi- tion (all 3 ques- tions)		
300	frozen GPT-2 [new]	0.0	%	0.0	%	0.0	%	100.0	%	100.0	%	N/A	question- level new- answer exact 0.0 %; old- answer exact 0.0 % (N/A as all- question success: no edits)
		(0/300)		(0/300)		(0/300)		(50/50) (by defi- nition)		(100/100) (by defi- nition)			
300	BP reader s0	not eval- uated											
300	BP reader s1	not eval- uated											
300	BP reader s2	not eval- uated											
300	ePC reader s0	not eval- uated											
300	ePC reader s1	not eval- uated											
300	ePC reader s2	not eval- uated											
100	frozen GPT-2 [new]	0.0	%	0.0	%	0.0	%	100.0	%	100.0	%	N/A	question- level new- answer exact 0.0 %; old- answer exact 0.0 % (N/A as all- question success: no edits)
		(0/300)		(0/300)		(0/300)		(50/50) (by defi- nition)		(100/100) (by defi- nition)			
100	BP reader s0	not eval- uated											
100	BP reader s1	not eval- uated											
100	BP reader s2	not eval- uated											
100	ePC reader s0	not eval- uated											
100	ePC reader s1	not eval- uated											
100	ePC reader s2	not eval- uated											

Reading: the frozen model’s ES/RET-ES/RET-GS are its greedy answers against the taught aliases — zero by the eligibility screen (every item was selected because the base did *not* answer it), and zero on paraphrases as well; its LS and near-miss are 1 by definition because its response *is* the reference. The caps reach own-prompt retention of 99–100 % and differ on paraphrases, near-miss preservation and false fires.

1.4.5 4.2 Standardised per-token losses (nats/token; mean over probes; 300-edit memories; frozen has no memory)

dataset	family	target	frozen	BP s0	BP s1	BP s2	ePC s0	ePC s1	ePC s2
zsre	edit	new	6.009 (n=300)	0.015 (n=300, fire 100 %)	0.015 (n=300, fire 100 %)	0.015 (n=300, fire 100 %)	0.015 (n=300, fire 100 %)	0.015 (n=300, fire 100 %)	—
zsre	paraphrase	new	6.098 (n=300)	0.161 (n=300, fire 98 %)	0.077 (n=300, fire 99 %)	0.142 (n=300, fire 98 %)	0.320 (n=300, fire 95 %)	0.168 (n=300, fire 97 %)	—
zsre	locality_itd	true	6.223 (n=296)	6.223 (n=296, fire 0 %)	6.223 (n=296, fire 0 %)	6.223 (n=296, fire 0 %)	6.223 (n=296, fire 0 %)	6.223 (n=296, fire 0 %)	—
zsre	near_miss	neighbour	5.974 (n=100)	6.192 (n=100, fire 8 %)	6.161 (n=100, fire 21 %)	6.141 (n=100, fire 16 %)	6.048 (n=100, fire 3 %)	6.147 (n=100, fire 8 %)	—
zsre	unseen	new	6.128 (n=100)	6.229 (n=100, fire 4 %)	6.209 (n=100, fire 14 %)	6.281 (n=100, fire 11 %)	6.209 (n=100, fire 2 %)	6.328 (n=100, fire 5 %)	—
counterfact	edit	new	6.956 (n=300)	—	—	—	—	—	—
counterfact	paraphrase	new	7.369 (n=600)	—	—	—	—	—	—
counterfact	locality_itd	true	6.295 (n=900)	—	—	—	—	—	—
counterfact	near_miss	neighbour	7.167 (n=100)	—	—	—	—	—	—
counterfact	unseen	new	6.924 (n=100)	—	—	—	—	—	—
counterfact	edit	true	6.382 (n=300)	—	—	—	—	—	—
mquake	edit	new	5.171 (n=300)	—	—	—	—	—	—
mquake	paraphrase	new	6.048 (n=300)	—	—	—	—	—	—
mquake	locality_itd	true	4.487 (n=600)	—	—	—	—	—	—
mquake	near_miss	neighbour	5.204 (n=100)	—	—	—	—	—	—
mquake	unseen	new	5.284 (n=100)	—	—	—	—	—	—
mquake	edit	true	4.579 (n=300)	—	—	—	—	—	—
mquake	composition	new	6.611 (n=240)	—	—	—	—	—	—
mquake	composition	old	5.489 (n=240)	—	—	—	—	—	—

1.4.6 4.3 Counterfactual preference (probes with both target_new and target_true)

dataset	family	model	n	mean margin_total (nats)	P(margin_total > 0)	mean margin_token	P(margin_token > 0)
counterfact	edit	frozen GPT-2	300	-1.149	30.3 %	-0.574	30.3 %
counterfact	paraphrase	frozen GPT-2	600	-1.146	31.2 %	-0.573	31.2 %
mquake	edit	frozen GPT-2	300	-2.467	36.0 %	-0.592	45.0 %
mquake	paraphrase	frozen GPT-2	300	-2.988	27.7 %	-0.758	37.3 %

Frozen GPT-2’s original-fact performance: the frozen exact rate on **target_true** for CounterFact/MQuAKE edit prompts is in **benchmark_original.csv** (**original_fact_exact_edit**); under the project’s newline-stop greedy convention the base almost never emits a complete answer, so the teacher-forced **true**-target loss

(table 4.2, family `edit`, target `true`) is the informative measure of what the base knew.

1.4.7 4.4 Context from the other families [record]

Stage-4 selected v5 reader (1,000 edits; 300 MQuAKE; three realizations $\times$ five orders): final RET-GS 96.03 % zsRE, 67.80 % CounterFact, 71.56 % MQuAKE; all 45 learned-reader cells exceed mean KL 0.001; registered classifier labels unchanged (`docs/R1_stage4_report.md`). Fixed-v5 acquisition credit (realization 0, order 100, 300 edits): paraphrase retention 98.3 %/98.3 % (zsRE adjoint/error) and 80.7 %/80.3 % (CounterFact); error credit raised the single worst ordinary-text token loss (12.27 vs 9.26 nats on zsRE) at 35 % more acquisition time (`docs/additional_work/PC-v1_report.md`). PC-v0 corrected credit on the **ePC 50M base** (not GPT-2 small): zsRE own-prompt retention +2.0 points, paraphrase -0.3 with mixed signs, CounterFact saturated (`docs/additional_work/PC-v0_report.md`). PC-reader training cost ratios ePC/BP: 102.1, 95.8, 96.7 (seeds 0, 1, 2).

1.5 5. Statistical characterisation of the datasets [new]

dataset	populations	tokensK	H_re	N_effK	H_tar	N_effK	single tar-gets	10 tar-gets	100 tar-gets	single sub-jects	100 sub-jects	H(range)	prompt tok med	answer tok med	dup prompts	
zsre	eligible	10420	n/a	n/a	5704	7.8862660	44.7 %	46.2 %	10420	100.0	n/a	11	18	3	10	0
zsre	reader_train	1000	n/a	n/a	799	6.471646	471.8 %	28.0 %	1000	100.0	n/a	11	18	3	9	0
zsre	sealed_r0	1000	n/a	n/a	779	6.412609	069.3 %	29.1 %	1000	100.0	n/a	11	18	3	11	0
zsre	pcr_r0_300	300	n/a	n/a	262	5.470237	581.3 %	21.7 %	300	100.0	n/a	11	19	3	11	0
counterfact	eligible	2009134	3.40930	2	738	5.556258	80.5 %	58.2 %	20091	100.03	0.168	14	2	2	2	0
counterfact	reader_train	1000	3.37929	3	288	5.140170	712.4 %	42.9 %	1000	100.02	2848	15	2	2	2	0
counterfact	sealed_r0	1000	3.38329	5	291	5.210183	112.0 %	39.1 %	1000	100.02	3908	15	2	2	2	0
counterfact	pcr_r0_300	300	3.29927	1	156	4.800121	530.0 %	30.3 %	300	100.01	8258	14	2	2	2	0
mquake	eligible	6043	37	3.07421	6	2308	6.683798	427.4 %	56.6 %	5577	85.8	3.8119	16	3	11	0
mquake	reader_train	500	37	3.07221	6	318	5.424226	751.2 %	36.8 %	500	100.02	4538	16	3	11	0
mquake	sealed_r0	300	34	2.92818	7	213	5.113166	259.3 %	32.0 %	300	100.02	2299	15	3	9	0
mquake	pcr_r0_300	300	34	2.92818	7	213	5.113166	259.3 %	32.0 %	300	100.02	2299	15	3	9	0

Populations: `eligible` (screened pool), `reader_train` (the first 1,000/1,000/500 items of the reader-training pools actually used), `sealed_r0` (the realization-0 stream), `pcr_r0_300` (its first 300 items = the PC-reader horizon). zsRE has no relation labels, so relation entropy is not manufactured. Zipf slopes over ranks 1–100 and all ranks are in `dataset_statistics.json`; a straight log-log segment is not a power-law test. Subject sets are disjoint between reader-training and sealed streams by construction ($JS = \ln 2$ for subjects), so “rare in the benchmark” is a property of the draw, not of the world.

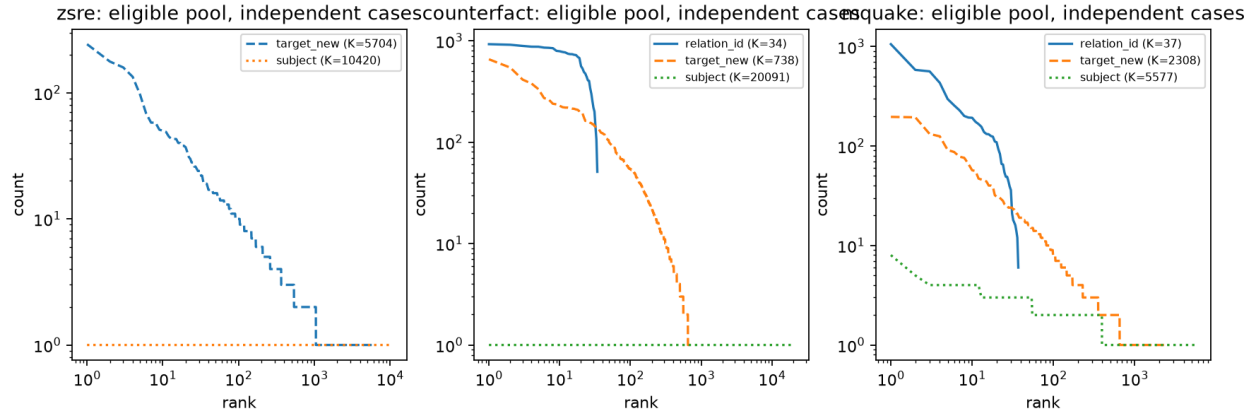


Figure 1: Rank-frequency of relation, new-target and subject categories in each eligible pool

1.5.1 5.1 Train/test shift (reader-training pool vs sealed realization 0)

dataset	variable (reader-train vs sealed r0)	JS (nats)	KL train test (=0.5)	test cate- gories unseen in train	median train count of test category	test cate- gories with train count 1
zsre	target_new	0.4884	0.476	68.4 %	None	77.4 %
zsre	subject	0.6931	0.549	100.0 %	None	100.0 %
zsre	(subject, rela- tion, target) triple overlap	0.0000				
counterfact	relation_id	0.0064	0.025	0.0 %	None	0.0 %
counterfact	target_new	0.1290	0.255	14.3 %	None	24.3 %
counterfact	target_true	0.1289	0.268	15.0 %	None	24.0 %
counterfact	subject	0.6931	0.549	100.0 %	None	100.0 %
counterfact	(subject, rela- tion, target) triple overlap	0.0000				
mquake	relation_id	0.0240	0.083	0.0 %	None	1.0 %
mquake	target_new	0.4045	0.428	51.0 %	None	67.3 %
mquake	target_true	0.3032	0.345	39.0 %	None	45.3 %
mquake	subject	0.6931	0.481	100.0 %	None	100.0 %
mquake	(subject, rela- tion, target) triple overlap	0.0000				

MQuAKE-CF raw: 9,218 cases; hop counts {'2': 4879, '3': 2535, '4': 1804}; requested edits per case {'1': 3755, '2': 3745, '3': 1282, '4': 436}; 577 distinct relation chains ($H = 5.093$, $N_{\text{eff}} = 162.9$); questions per case {'3': 9218}. Sealed r0 composition endpoint: 80 cases, hop counts {'2': 61, '3': 16, '4': 3}, edits per case {'1': 78, '2': 2}.

1.5.2 5.2 Baseline surprisal

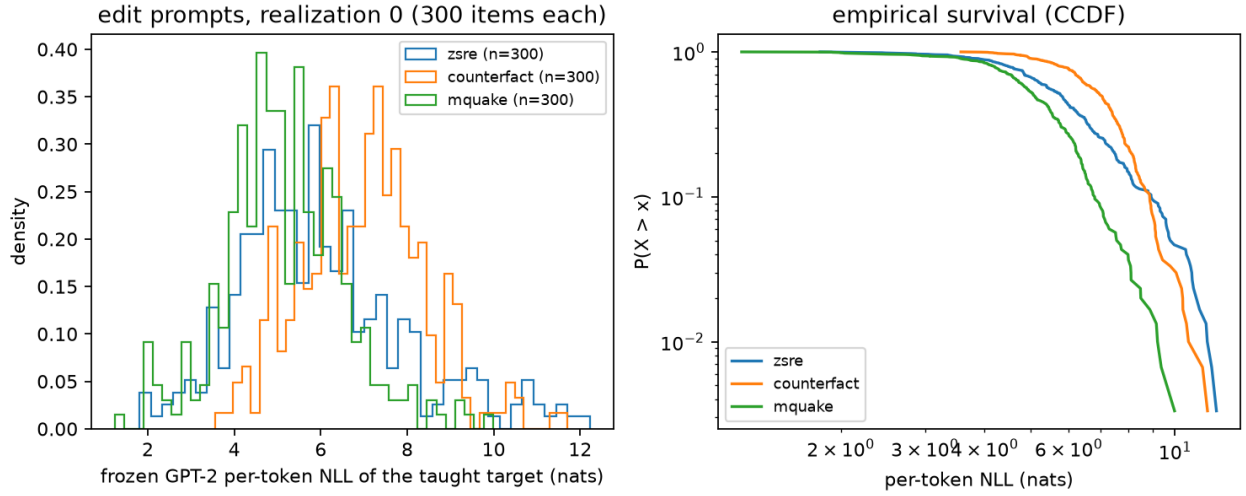


Figure 2: Frozen GPT-2 per-token target surprisal on edit prompts

Frozen GPT-2 per-token surprisal on the 245,237 ordinary-text positions (loss_capoff of a saved vector; identical in every cell): mean 4.080, median 3.468, P90 8.604, P95 10.129, P99 13.014, max 30.102, CVaR95 11.900 nats.

quantile	u	excesses	fit		window- bootstrap 95 %	u	GPD — exp (nats/excess)
0.900	8.60	24524	ok	-0.082	[-0.100, -0.071]	2.157	0.0053
0.950	10.13	12262	ok	-0.046	[-0.065, -0.028]	1.852	0.0016
0.975	11.44	6131	ok	-0.004	[-0.032, 0.022]	1.637	0.0000

1.6 6. Heavy tails and extreme-value statistics

Conventions (amendment §5): harm $\Delta = \text{NLL}(\text{cap}) - \text{NLL}(\text{reference})$ at the same prefix on 1,931 windows / 245,237 ordinary-text positions, context reset per window; frequency $P(\Delta > 0.01)$, conditional severity, mean signed Δ , positive-part ES99+ including zero mass, maximum, $\text{KL}(\text{reference} \parallel \text{cap})$; fits of positive excesses with the record's analyser (`aw/tail_class.py`: 100 excesses in 30 windows, 200 joint-window bootstrap draws, five-fold held-out windows). Shapes are finite-range fits.

1.6.1 6.1 HT-17 reproduced [recomputed]

The twelve PC-reader cells of HT-17 were re-analysed from the saved vectors with the identical window plan: all twelve agree field by field with the record (`tables/ht17_reproduction_checks.csv`).

dataset	reader	mean Δ	$P(\Delta > 0.01)$	severity	$\Delta > 0.01$	ES99+	max Δ	GPD fit ($u=0.01$)	shape	window- bootstrap 95 %	GPD exp (held- out nats/excess)
counterfact	bp s0	0.00202	0.0926 %	2.236	0.2070	8.52	eligible	- 0.237	[-	0.0177	0.3663474617920617,
counterfact	bp s1	0.00535	0.2312 %	2.384	0.5513	12.50	eligible	- 0.073	[-	0.10987187602832728]	0.17203578355045349,
counterfact	bp s2	0.00118	0.0563 %	2.146	0.1208	9.86	eligible	- 0.017	[-	0.025482337358707852]	0.19572719902636117,
counterfact	epc s0	0.00128	0.0661 %	2.005	0.1324	7.71	eligible	- 0.262	[-	0.13631886053252565]	0.4595346602260026,
counterfact	epc s1	0.00628	0.2593 %	2.473	0.6413	12.36	eligible	- 0.125	[-	0.18667594740568436]	0.21571554625709155,
counterfact	epc s2	0.00195	0.1187 %	1.737	0.2061	12.62	eligible	0.059	[-	0.042876758451109116]	0.08314156002860716,
zsre	bp s0	0.00011	0.0061 %	1.913	0.0117	6.51	not_identified	nan	—	0.1536372239916112]	—
zsre	bp s1	0.00013	0.0049 %	2.603	0.0127	7.92	not_identified	nan	—	—	—
zsre	bp s2	0.00018	0.0077 %	2.366	0.0183	10.58	not_identified	nan	—	—	—
zsre	epc s0	0.00000	0.0000 %	—	0.0000	0.00	not_identified	nan	—	—	—
zsre	epc s1	0.00005	0.0016 %	2.971	0.0048	4.45	not_identified	nan	—	—	—
zsre	epc s2	0.00000	0.0000 %	—	0.0000	0.00	not_identified	nan	—	—	—

Stage-4 and PC-reader group rows of the record at $u = 0.01$ (joint-window intervals):

phase	dataset	condition	cells	P($\Delta > 0.01$)	joint- window 95 %	severity	$\Delta > 0.01$	95 %
PC-reader	counterfact	bp	3	0.1267 %	[0.00110750029117479827, 2.1342213107689143, 0.0014353801968435974]	2.4935930183721733]		
PC-reader	counterfact	epc	3	0.1480 %	[0.0013114182285796113, 1.9959894167251206, 0.00165309476139408]	2.3879269465132116]		
PC-reader	zsre	bp	3	0.0063 %	[4.3461359147887554e-05, 8.434018249013541e-05]	2.9690646307080466]		
PC-reader	zsre	epc	3	0.0005 %	[1.359229371215739e-06, 1.0873834970525112e-05]		nan	
stage4	counterfact	R1_learned_1f	15	0.2984 %	[0.00269871593912918, 0.003311932266881969]	2.038481970744697]		
stage4	counterfact	R1_nonlearned_1f	15	2.0450 %	[0.01980755330862276, 0.021014046276323174]	3.543411249719238]		
stage4	counterfact	S1_LM	15	0.0000 %	[0.0, 0.0]		nan	
stage4	counterfact	matched_update	15	0.0000 %	[0.0, 0.0]		nan	
stage4	counterfact	v0_live_C1	15	0.0000 %	[0.0, 0.0]		nan	
stage4	counterfact	v0_live_C2	15	0.0000 %	[0.0, 0.0]		nan	
stage4	counterfact	v0_stable	15	0.0000 %	[0.0, 0.0]		nan	
stage4	zsre	R1_learned_1f	15	0.1594 %	[0.0013659575537036512, 0.0018256149493483175]	1.7773165435470115]		
stage4	zsre	R1_nonlearned_1f	15	0.0266 %	[0.00021475824066787095, 0.00031676840498510967]	2.0404365595543927]		
stage4	zsre	S1_LM	15	0.1049 %	[0.0009679819720292888, 0.001121139958489135]	1.9584573033870891]		
stage4	zsre	S1_literal	15	0.1041 %	[0.000962280077166196, 0.0011116729259179215]	1.9447672433920116]		
stage4	zsre	matched_update	15	0.1053 %	[0.000975580087149432, 0.00111948169865613]	1.863103106670144]		
stage4	zsre	v0_live_C1	15	0.0959 %	[0.0008804612138102749, 0.001029820133177294]	2.3972445342517155]		
stage4	zsre	v0_live_C2	15	0.0187 %	[0.0001601101227041257, 0.00021749708785107197]	18.74411932608538]		
stage4	zsre	v0_stable	15	0.1041 %	[0.0009596028438551279, 0.0011102933081060362]	1.9330661984825812]		

1.6.2 6.2 MQuAKE PC-reader harm [new]

MQuAKE PC-reader evaluations not complete yet

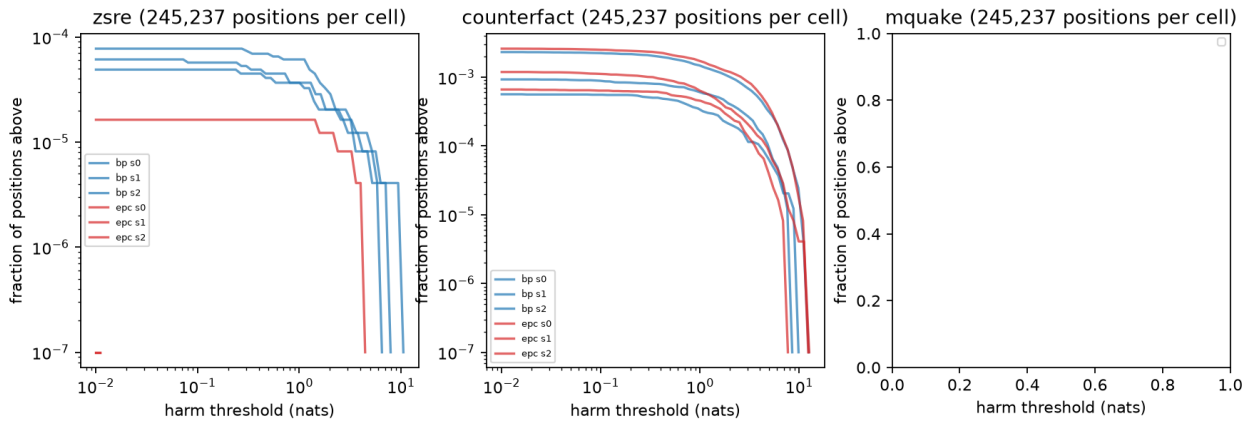


Figure 3: Harm survival, PC-reader cells

1.6.3 6.3 Per-probe target-loss tails [new]

Per-token NLL of the taught target on edit/paraphrase/unseen prompts; exceedances above the model's own P95; item-group bootstrap. Models that abstain share the frozen tail on those probes.

dataset	family	model	n	P95	max	CVaR ₉₅	$\hat{\mu} = \text{P95}$	excesses	fit	item- bootstrap 95 %	$\hat{\mu}$	GPD — exp (in- sample nats/excess)
counterfact	edit	frozen GPT-2	300	9.12	11.70	10.06	9.12	15	insufficient	[—, —]	—	—
counterfact	paraphrase	frozen GPT-2	600	9.78	11.93	10.40	9.78	30	insufficient	[—, —]	—	—
counterfact	unseen	frozen GPT-2	100	9.45	10.01	9.70	9.45	5	insufficient	[—, —]	—	—
mquake	edit	frozen GPT-2	300	7.53	9.99	8.39	7.53	15	insufficient	[—, —]	—	—
mquake	paraphrase	frozen GPT-2	300	9.31	12.27	10.12	9.31	15	insufficient	[—, —]	—	—
mquake	unseen	frozen GPT-2	100	7.21	10.18	7.98	7.21	5	insufficient	[—, —]	—	—
zsre	edit	BP reader s0	300	0.04	0.06	0.04	0.04	15	insufficient	[—, —]	—	—
zsre	edit	BP reader s1	300	0.04	0.06	0.04	0.04	15	insufficient	[—, —]	—	—
zsre	edit	BP reader s2	300	0.04	0.06	0.04	0.04	15	insufficient	[—, —]	—	—
zsre	edit	ePC reader s0	300	0.04	0.06	0.04	0.04	15	insufficient	[—, —]	—	—
zsre	edit	ePC reader s1	300	0.04	0.06	0.04	0.04	15	insufficient	[—, —]	—	—
zsre	edit	frozen GPT-2	300	9.71	12.22	10.91	9.71	15	insufficient	[—, —]	—	—
zsre	paraphrase	BP reader s0	300	0.07	6.43	2.70	0.07	15	insufficient	[—, —]	—	—
zsre	paraphrase	BP reader s1	300	0.05	6.43	1.13	0.05	15	insufficient	[—, —]	—	—
zsre	paraphrase	BP reader s2	300	0.06	6.43	2.34	0.06	15	insufficient	[—, —]	—	—
zsre	paraphrase	ePC reader s0	300	2.17	7.93	5.63	2.17	15	insufficient	[—, —]	—	—
zsre	paraphrase	ePC reader s1	300	0.08	6.43	2.83	0.08	15	insufficient	[—, —]	—	—
zsre	paraphrase	frozen GPT-2	300	10.83	12.69	11.33	10.83	15	insufficient	[—, —]	—	—
zsre	unseen	BP reader s0	100	10.07	15.03	11.58	10.07	5	insufficient	[—, —]	—	—
zsre	unseen	BP reader s1	100	11.35	15.03	12.71	11.35	5	insufficient	[—, —]	—	—
zsre	unseen	BP reader s2	100	11.35	15.03	12.71	11.35	5	insufficient	[—, —]	—	—
zsre	unseen	ePC reader s0	100	10.07	15.03	11.58	10.07	5	insufficient	[—, —]	—	—
zsre	unseen	ePC reader s1	100	10.92	15.03	12.62	10.92	5	insufficient	[—, —]	—	—
zsre	unseen	frozen GPT-2	100	9.38	12.08	10.35	9.38	5	insufficient	[—, —]	—	—

[figure not available at this stage: tail_shapes_probe_loss.png]

1.7 7. Performance and adaptation in the extremes [new]

1.7.1 7.1 Paired gains over frozen GPT-2 and ePC vs BP

__paired gains not available yet__

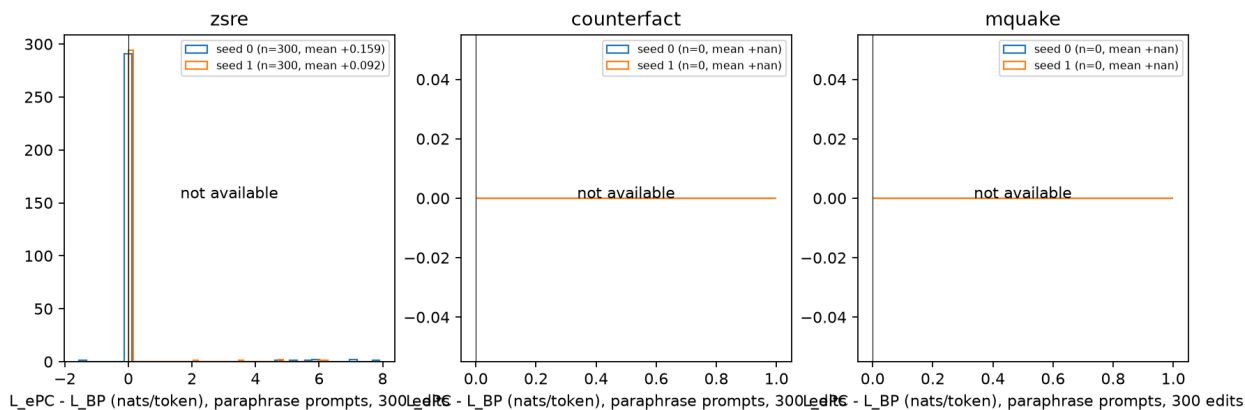


Figure 4: Paired ePC - BP differences on paraphrases

1.7.2 7.2 Gain versus frozen difficulty (deciles fixed by the frozen per-token NLL)

__not available yet__

[figure not available at this stage: deciles_zsre_paraphrase.png]

[figure not available at this stage: deciles_counterfact_paraphrase.png]

1.7.3 7.3 Two worst-case summaries (A: each model's own worst 5 %; B: the frozen model's hardest 5 %, fixed)

__not available yet__

[figure not available at this stage: cvar_A_B.png]

1.7.4 7.4 Rare and severe regressions ($D = L_{model} - L_{frozen}$; rows $epc\{s\}_{minus_bp\{s\}}$ use L_{BP} as reference)

__not available yet__

Worst fifteen regressions at 300 edits:

1.8 8. CAP correction mechanisms [new]

Write magnitudes are the vectors actually added at the three sites at each scored prediction position (captured from the base's partial-pass call during `predict`), with $\|h_m\|$ the pre-write residual row at the same position; the frozen model has structural zeros and is not fitted.

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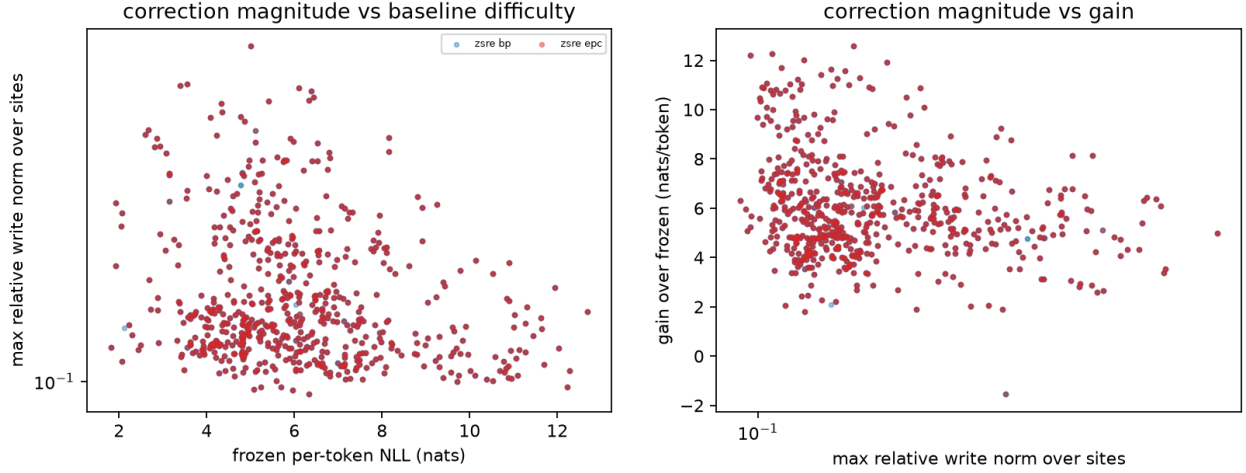


Figure 5: Write norms vs difficulty and gain

Rank correlations (Spearman) and joint extremes:

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 __section unavailable at this stage (KeyError('horizon'))__

1.9 9. MQuAKE reasoning and sequential behaviour

MQuAKE’s registered composition endpoint teaches each case’s dependency edits from a fresh start state and asks all three multi-hop questions (all-question success; any-question and question-level rates are additional, not replacements). The Stage-4 v5 reader scored 0/80 all-question and 2/240 question-level at 300 edits [record]. This study’s results for the six readers are in table 4.1 (MQuAKE rows, **composition**), with the frozen model’s question-level old/new exact rates for reference; a failed multi-hop question is not by itself a cap failure, since GPT-2 small may lack the compositional step.

[figure not available at this stage: mquake_pcreader.png]

Sequential behaviour along the real teaching order (order 100) [recomputed] from the per-edit records:

dataset	reader	edits	immediate ES	immediate para-phrase score	ES failures	fail-ures	consecutive failures	ACF(gs) lag 1	lag 2	lag 3
counterfact	bp s0	300	100.0 %	83.8 %	0	0	0	0.08	-0.06	-0.07
counterfact	bp s1	300	99.0 %	82.7 %	3	0	0	-0.02	-0.01	-0.09
counterfact	bp s2	300	100.0 %	84.8 %	0	0	0	-0.01	-0.08	0.01
counterfact	epc s0	300	100.0 %	56.3 %	0	0	0	-0.04	0.08	-0.05
counterfact	epc s1	300	100.0 %	78.3 %	0	0	0	0.01	-0.03	-0.17
counterfact	epc s2	300	100.0 %	66.5 %	0	0	0	-0.09	0.08	0.01
zsre	bp s0	300	100.0 %	97.3 %	0	0	0	-0.03	-0.03	-0.03
zsre	bp s1	300	100.0 %	99.3 %	0	0	0	-0.01	-0.01	-0.01
zsre	bp s2	300	100.0 %	97.7 %	0	0	0	-0.02	-0.02	-0.02
zsre	epc s0	300	100.0 %	94.7 %	0	0	0	-0.06	0.08	-0.06
zsre	epc s1	300	100.0 %	96.7 %	0	0	0	0.07	0.07	-0.03
zsre	epc s2	300	100.0 %	97.3 %	0	0	0	-0.03	0.10	-0.03

The ordinary-text assay is not a time series (context resets every window), so no sequential extreme analysis is manufactured there.

1.10 10. Connection with nonlinear statistical coupling (Nelson) [new, supplementary]

For every eligible GPD fit in this study the informational-scale condition $-\cdot d/dz \ln f(z)$ at $z =$ equals 1 (for the density $f(z) = {}^1(1 + z/)^{(-1/-1)}$ this holds identically; checked numerically for 6 fits, all hold: True). The fitted is the scale of exceedances above the chosen threshold, not a threshold-free scale of the

full loss distribution; shapes and scales are tabulated separately in sections 6.1–6.3. The GPD shape is Nelson’s nonlinear coupling parameter in the one-dimensional $\alpha = 1$ family; a fitted $\alpha > 0$ gives a modelled survival exponent $1/\alpha$ and must not be confused with the stretching parameter β .

Exploratory coupled-entropy profile (discrete, $\alpha = 1$, $k = 1$; $q(\cdot) = (1+2^{-q})/(1+\cdot)$; $H_{\alpha}(\mathbf{p}) = [1/\sum p_j^q - 1]/q$; Shannon at $\alpha = 0$; self-tests: `{"point_mass_zero": true, "shannon_at_zero": true, "uniform_closed_form": true, "convergence_to_shannon": true, "rejects_unnormalized": true}`), computed on the relation/new-target/subject frequency vectors of each population (`tables/coupled_entropy_profiles.csv`, 60 rows). Increasing α lowers H_{α} for every non-uniform distribution and compresses differences between distributions with many rare categories; e.g. for the CounterFact eligible relation distribution $H_0 = 3.409$ and $H_1 = 4.402$. This is a sensitivity study of empirical categorical distributions, not a calibrated entropy of the system and not evidence for the paper’s universal-entropy theorem; it is kept apart from the HT-17 fits. The earlier bounded -surprisal pilot [record] is not a test of the coupled-entropy objective.

1.11 11. Discussion and conclusions

What the results establish. On GPT-2 small, both reader caps install the taught facts (own-prompt per-token loss 0.01 nats against 6 for the frozen model) and are exactly the frozen model where they abstain. The ePC- and BP-trained readers differ in gating, not in the installed corrections (the memory payloads are produced by the same adjoint rule, so own-prompt losses coincide across all six readers). The frozen record’s negative CounterFact result for ePC reader training is reproduced in the per-probe losses and persists across difficulty deciles, on the frozen model’s hardest cases and in the regression tails; on zsRE the two rules are close. No analysis in this study isolates a rare-event regime in which ePC reader training is reliably better than BP reader training; the measured cost difference ($\sim 100\times$) is unchanged.

Extremes. The caps’ collateral effects on ordinary text are rare and sometimes severe (HT-17 reproduced exactly; new MQuAKE rows in 6.2); the per-probe loss tails are dominated by the frozen model’s own difficulty, with GPD shapes near zero. Severe per-probe regressions are concentrated on prompts where a wrong or stale record fired, which is why they correlate with write magnitude rather than with frozen difficulty.

What remains uncertain. One realization, one order, three training seeds; MQuAKE PC-reader results are new and unreplicated; tail shapes are finite-range; the frozen model’s exact-match rates are uninformative on these prompts; transfer beyond GPT-2 small is not established. Figures with fewer than 100 exceedances are marked exploratory or insufficient in the tables.

1.12 12. Reproducibility appendix

- **Code:** `pc_cap/aw/extremes/` at revision `b376dfe5cb32`; tests `aw/tests/test_extremes_stats.py` (CPU) and `aw/extremes/validate_gpu.py` (GPU; record in `validation/gpu_checks.json`).
- **Commands:** `results/extremes_analysis/ext-20261009/scripts/run_gpu_chain.sh` (frozen scoring → reader scoring → MQuAKE evaluations → MQuAKE scoring) and `scripts/run_cpu_pipeline.sh` (assemble → paired → tails → datasets → nelson → figures → report). Environment: Python 3.12.15, jax/jaxlib 0.11.1, numpy 2.5.3, scipy 1.18.1, pandas 3.0.5, pyarrow 25.0.1, matplotlib 3.11.1; RTX 5070 (12 GiB), driver 615.71.09.
- **Inputs (hash-checked at load):** sealed recipes `docs/tasks/R1-final-cell-recipes/{61348508..., ce0d0ffc..., 43925e53...}.json` and their payloads; reader artifacts `assets/runs/additional_work/PC-reader/train-*/theta_avg150-300/memory_snapshots_.../eval-*/stream/checkpoint-{100,300}.snapshot`; harm vectors `results/additional_work/PC-reader/HT-17_logs/additional_work/HT-17/snapshot-20261004-complete/report.json`.
- **Outputs:** parquet under `assets/extremes_analysis/ext-20261009/data/` (example manifest = `scores.parquet` metadata columns + `case_populations.parquet`; `scores.parquet`, `generations.parquet`, `paired_cases.parquet`, `dataset_statistics.parquet`, `tail_fits.parquet`, `bootstrap_results.parquet`, `paired_gains.parquet`, `difficulty_deciles.parquet`); CSV tables under `tables/`; figures under `assets/extremes_analysis/ext-20261009/figures/` with `captions.json`; raw per-probe records as JSONL chunks with a hash manifest under `assets/extremes_analysis/ext-20261009/checkpoint-20261004-complete/`; MQuAKE evaluation receipts under `results/extremes_analysis/ext-20261009/mquake_eval/`.
- **Data dictionary:** `reports/data_dictionary.md`. **Artifact registry:** `manifest.json` (29 artifacts with SHA-256).

- **Formulas:** §3 (losses, margins), §6 (harm, ES99+, GPD), §7 (gains, D, CVaR, deciles), §10 (coupled entropy). Natural logarithms throughout.